Experiment 2

Linear Regression

Aim:

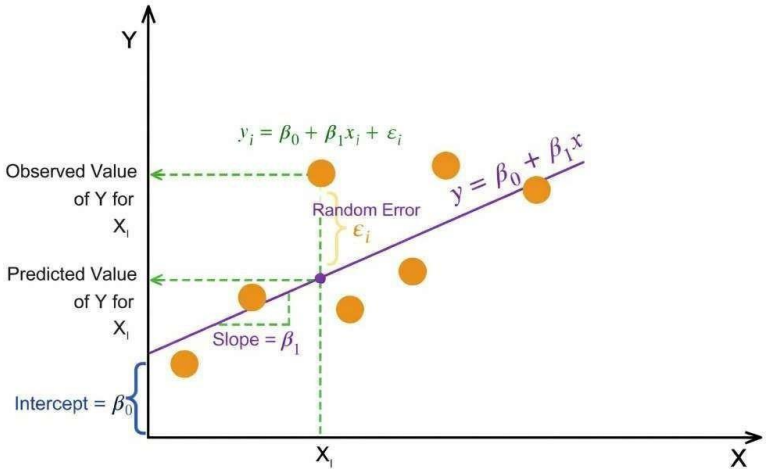
To implement and visualize simple linear regression and multiple linear regression, and to evaluate model performance using standard regression metrics.

Theory:

Linear Regression is a supervised machine learning algorithm used to model the relationship between a dependent (target) variable and one or more independent (feature) variables by fitting a linear equation to observed data. The objective is to predict continuous numerical outcomes.

Simple Linear Regression: involves a single independent variable and is mathematically expressed as:

$$Y = \beta_0 + \beta_1 X + \epsilon$$



Multiple Linear Regression: involves two or more independent variables and is expressed as:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n + \epsilon$$

Where,

- Y: Dependent variable (Target)
- X: Independent variable(s) (Features)
- β_0: Intercept

- $\beta_1, \beta_2, \ldots, \beta_n$: Regression coefficients
- ϵ: Error term (residual)

Method of Least Squares

The coefficients of the regression model are estimated using the **Least Squares Method**, which minimizes the **Residual Sum of Squares (RSS)**—the sum of squared differences between observed and predicted values.

Assumptions of Linear Regression

1. **Linearity** – The relationship between independent and dependent variables is linear.
2. **Independence** – Observations are independent of each other.
3. **Homoscedasticity** – Constant variance of residuals across all values of X.
4. **Normality** – Residuals are normally distributed.

Merits of Linear Regression

- **Simplicity and Interpretability:** Linear Regression is easy to understand and implement. Model coefficients have clear statistical meaning, indicating the magnitude and direction of feature influence on the target variable.
- **Effective for Linearly Related Data:** Performs well when the true relationship between features and target is approximately linear and assumptions are reasonably satisfied.

Demerits of Linear Regression

- **Linearity Assumption:** The model cannot capture non-linear relationships unless manual feature transformations (e.g., polynomial terms) are applied.
- **Sensitivity to Outliers:** Least Squares estimation squares residuals, giving excessive weight to outliers, which can significantly distort the model.
- **Multicollinearity Issues:** High correlation among independent variables leads to unstable coefficient estimates and reduced interpretability in Multiple Linear Regression.

Algorithm

Step 1: Let X be the input features and (Y) be output values

Step 2: Assume a linear relationship: $y = \beta_o + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$

- β_o is the intercept (value when all x = 0)
- $\beta_1, \beta_2, \ldots$ are coefficients (weights) for each feature

Step 3: Define the error (residual) for each data point:

- Error = Actual value - Predicted value
- $e_i = y_i - (\beta_o + \beta_1 x_{1i} + \beta_2 x_{2i} + \ldots)$

Step 4: Calculate total error using Sum of Squared Errors (SSE):

- $SSE = \Sigma(e_i)^2 = \Sigma(y_i - \hat{y}_i)^2$

Step 5: Find coefficients that minimize SSE using Normal Equation:

- $\beta = (X^T X)^{-1} X^T y$
- Where X is the feature matrix and y is the target vector

Step 6: For new data, predict using:

- $\hat{y} = \beta_o + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_n x_n$

Linear Regression - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Option A: Single Variable Regression

Step 1: Importing Libraries

 # Importing Libraries

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
print("Libraries Imported");
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

 # Read and store the salary dataset

```
data = pd.read_csv('Salary_dataset.csv')
print("Dataset reading complete")
```

Output:

```
Dataset reading complete
```

Step 3: Data Analysis

 # Display the first 5 rows of the dataset

```
data.head()
```

Output:

index	YearsExperience	Salary
0	1.2	39344
1	1.4	46206
2	1.6	37732
3	2.1	43526
4	2.3	39892

Display the last 5 rows of the dataset

```
data.tail()
```

Output:

index	YearsExperience	Salary
25	9.1	105583
26	9.6	116970
27	9.7	112636
28	10.4	122392
29	10.6	121873

Summary statistics for numerical columns

```
data.describe()
```

Output:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.413333	76004.000000
std	2.837888	27414.429785
min	1.200000	37732.000000
25%	3.300000	56721.750000
50%	4.800000	65238.000000
75%	7.800000	100545.750000
max	10.600000	122392.000000

Concise summary of a DataFrame

```
data.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
#    Column            Non-Null Count   Dtype
---  ---------          ----------------  -------
0    YearsExperience   30 non-null      float64
1    Salary            30 non-null      int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

Detect missing values

```
data.isnull().sum()
```

Output:

```
YearsExperience    0
Salary             0
dtype: int64
```

 # Return a tuple representing the dimensionality of the DataFrame

```
data.shape
```

Output:

```
(30, 2)
```

Step 4: Data Preprocessing

 # Convert YearsExperience to the nearest lower integer by removing decimal values

```
data['YearsExperience'] = np.floor(data['YearsExperience']).astype(int)
data
```

Output:

```
Dataset after Preprocessing:
```

Output:

index	YearsExperience	Salary
0	1	39344
1	1	46206
2	1	37732
3	2	43526
4	2	39892
5	3	56643
6	3	60151
7	3	54446
8	3	64446
9	3	57190
10	4	63219
11	4	55795
12	4	56958
13	4	57082
14	4	61112

... 15 more rows (total: 30 rows)

Step 5: Model Training

 # Create feature matrix X and target vector Y

```
X = data.drop(['Salary'], axis=1)
print("Feature matrix X created.")
```

Output:

```
Feature matrix X created.
```

Display the first 5 rows of the feature matrix X

```
X.head()
```

Output:

	YearsExperience
0	1
1	1
2	1
3	2
4	2

Create target vector Y

```
Y = data['Salary']
print("Target vector Y created.")
```

Output:

```
Target vector Y created.
```

Display the first 5 rows of the target vector Y

```
Y.head()
```

Output:

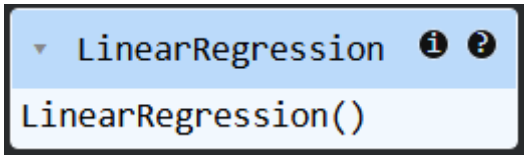
```
0    39344
1    46206
2    37732
3    43526
4    39892
Name: Salary, dtype: int64
```

Split the dataset into training and testing sets

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.15, random_state =
    123)
model = LinearRegression()
model.fit(X_train, Y_train)
print("Model Trained")
```

Output:

```
Model Trained
```



Step 6: Model Evaluation

 # Generate predictions for training and testing datasets using the trained model

```
Y_train_pred = model.predict(X_train)
Y_test_pred = model.predict(X_test)
print("Predictions generated for both Train and Test sets.")
```

Output:

```
Predictions generated for both Train and Test sets.
```

 # Evaluate model performance on training data using MAE, MSE, RMSE, and R2 score

```
mae = mean_absolute_error(Y_train, Y_train_pred)
mse = mean_squared_error(Y_train, Y_train_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_train, Y_train_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root MSE (RMSE):", rmse)
print("R2 Score:", r2)
```

Output:

```
Mean Absolute Error (MAE): 5457.139011764704
Mean Squared Error (MSE): 39230345.16254117
Root MSE (RMSE): 6263.413219845963
R2 Score: 0.9422825046716273
```

 # Evaluate model performance on testing data using MAE, MSE, RMSE, and R2 score

```
mae = mean_absolute_error(Y_test, Y_test_pred)
mse = mean_squared_error(Y_test, Y_test_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_test, Y_test_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root MSE (RMSE):", rmse)
print("R2 Score:", r2)
```

Output:

```
Mean Absolute Error (MAE): 3245.035764705882
Mean Squared Error (MSE): 20356724.219709344
Root MSE (RMSE): 4511.842663447978
R2 Score: 0.9774778470484203
```

 # Retrieve the learned coefficient (slope) of the linear regression model

```
model.coef_
```

Output:

```
array([9704.95294118])
```

 # Retrieve the intercept (bias term) of the linear regression model

```
model.intercept_
```

Output:

```
np.float64(26104.915294117643)
```

 # Plot actual training data points and the fitted linear regression line for visualization

```
plt.scatter(X_train, Y_train, label='Actual Salary Data')
plt.plot(X_train, Y_train_pred, color='red', label='Fitted Regression Line')
plt.xlabel('Years of Experience'); plt.ylabel('Salary')
plt.title('Linear Regression Fit')
plt.legend(); plt.show()
```

Output:

```
Linear Regression Fit (Training Data)
```

 # Plot actual testing data points and the fitted linear regression line to evaluate model performance

```
plt.scatter(X_test, Y_test, label='Actual Salary Data')
plt.plot(X_test, Y_test_pred, color='red', label='Fitted Regression Line')
plt.xlabel('Years of Experience'); plt.ylabel('Salary')
plt.title('Linear Regression Fit')
plt.legend(); plt.show()
```

Output:

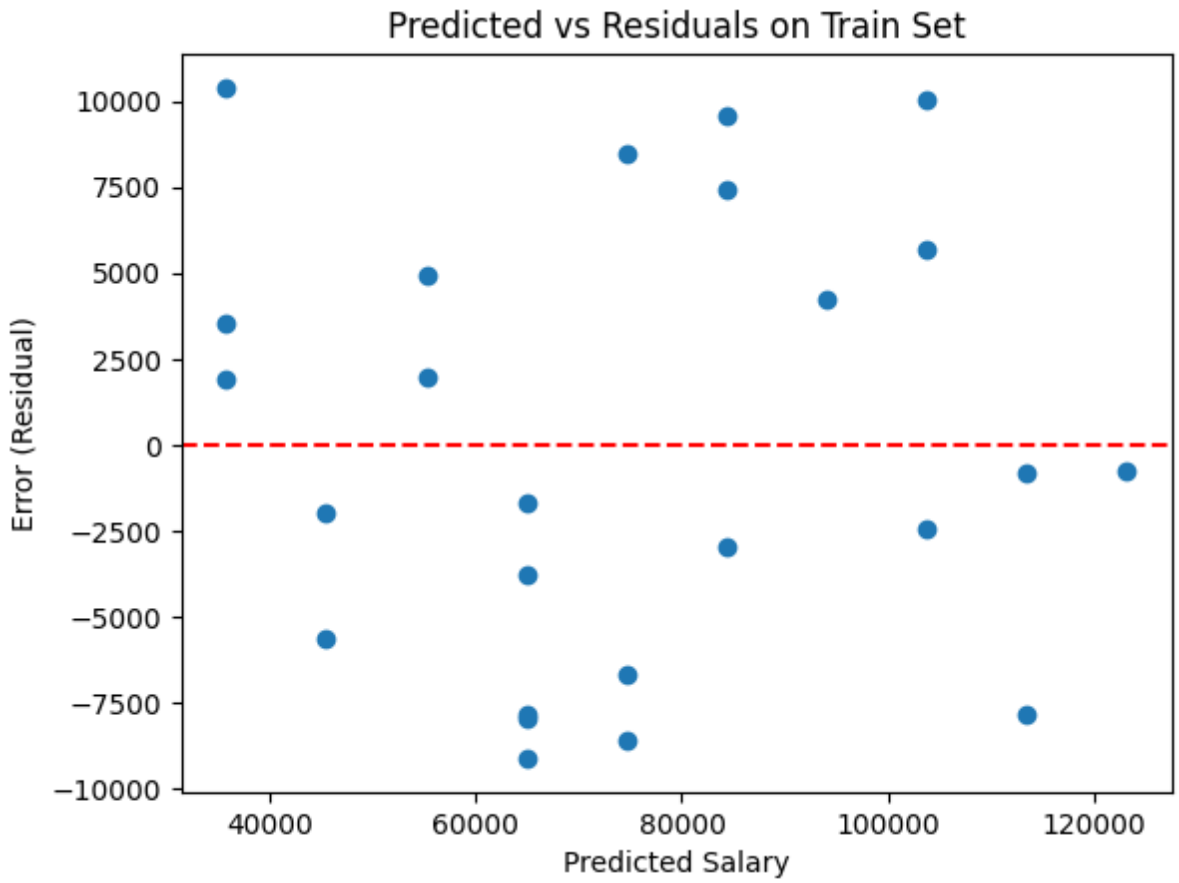
```
Linear Regression Fit (Testing Data)
```

 # Plot residuals against predicted values on train data to check error patterns and model assumptions

```
residuals = Y_train - Y_train_pred
plt.scatter(Y_train_pred, residuals); plt.axhline(y=0, color='red', linestyle='--')
plt.xlabel("Predicted Salary")
plt.ylabel("Error (Residual)")
plt.title("Predicted vs Residuals on Train Set"); plt.show()
```

Output:

```
Predicted vs Residuals (Train Set)
```

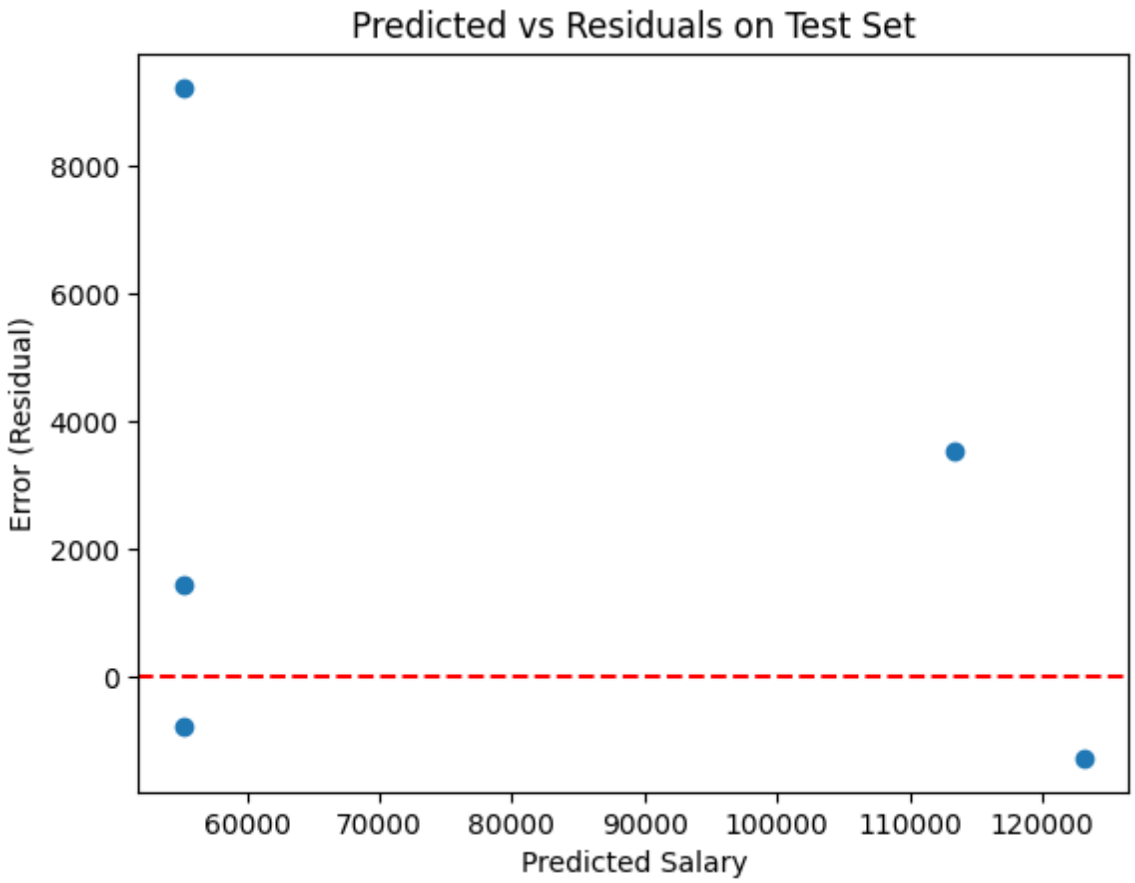


 # Plot residuals versus predicted values on test data to assess model generalization and error distribution

```
residuals = Y_test - Y_test_pred
plt.scatter(Y_test_pred, residuals)
plt.axhline(y=0, color='red', linestyle='--') # reference line for zero error
plt.xlabel("Predicted Salary")
plt.ylabel("Error (Residual)")
plt.title("Predicted vs Residuals on Test Set")
plt.show()
```

Output:

```
Predicted vs Residuals (Test Set)
```



 # Predict test values, compare actual vs predicted results, and display the top 5 most accurate predictions

```
Y_test_pred = model.predict(X_test)

comparison_df = pd.DataFrame({
    "Actual Salary": Y_test,
    "Predicted Salary": np.round(Y_test_pred, 2),
    "Error": np.round(Y_test - Y_test_pred, 2),
    "Absolute Error": np.round(abs(Y_test - Y_test_pred), 2)
})

best_predictions = comparison_df.sort_values(by="Absolute Error").head(5)
print(best_predictions)
```

Output:

```
Best Predictions:
```

Output:

	Actual Selling Price	Predicted Selling Price	Error	Absolute Error
7	54446	55219.77	-773.77	773.77
29	121873	123154.44	-1281.44	1281.44
5	56643	55219.77	1423.23	1423.23
26	116970	113449.49	3520.51	3520.51
8	64446	55219.77	9226.23	9226.23

Option B: Multi-Variable Regression

Step 1: Importing Libraries

 # Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
print("Libraries Imported")
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

 # Read and store the Car dataset

```
data = pd.read_csv('car_data.csv')
print("Dataset reading complete")
```

Output:

```
Dataset reading complete
```

Step 3: Data Analysis

 # Display specific selected rows from the dataset using their index positions

```
rows_to_show = [31, 144, 42, 114, 89]
data.loc[rows_to_show]
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
31	ritz	2011	2.35	4.89	54200	Petrol	Dealer	Manual	0
144	Bajaj Pulsar NS 200	2014	0.60	0.99	25000	Petrol	Individual	Manual	0
42	sx4	2008	1.95	7.15	58000	Petrol	Dealer	Manual	0
114	Royal Enfield Classic 350	2015	1.15	1.47	17000	Petrol	Individual	Manual	0
89	etios g	2014	4.75	6.76	40000	Petrol	Dealer	Manual	0

 # Display the first 5 rows of the dataset

```
data.head()
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	Petrol	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	Diesel	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	Petrol	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	Petrol	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	Diesel	Dealer	Manual	0

Display the last 5 rows of the dataset

```
data.tail()
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
296	city	2016	9.50	11.6	33988	Diesel	Dealer	Manual	0
297	brio	2015	4.00	5.9	60000	Petrol	Dealer	Manual	0
298	city	2009	3.35	11.0	87934	Petrol	Dealer	Manual	0
299	city	2017	11.50	12.5	9000	Diesel	Dealer	Manual	0
300	brio	2016	5.30	5.9	5464	Petrol	Dealer	Manual	0

Show statistical summary of numerical columns

```
data.describe()
```

Output:

	Year	Selling_Price	Present_Price	Kms_Driven	Owner
count	301.000000	301.000000	301.000000	301.000000	301.000000
mean	2013.627907	4.661296	7.628472	36947.205980	0.043189
std	2.891554	5.082812	8.644115	38886.883882	0.247915
min	2003.000000	0.100000	0.320000	500.000000	0.000000
25%	2012.000000	0.900000	1.200000	15000.000000	0.000000
50%	2014.000000	3.600000	6.400000	32000.000000	0.000000
75%	2016.000000	6.000000	9.900000	48767.000000	0.000000
max	2018.000000	35.000000	92.600000	500000.000000	3.000000

Display dataset structure and data types

```
data.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 301 entries, 0 to 300
Data columns (total 9 columns):
```

Output:

#	Column	Non-Null Count	Dtype
0	Car_Name	301 non-null	object
1	Year	301 non-null	int64
2	Selling_Price	301 non-null	float64
3	Present_Price	301 non-null	float64
4	Kms_Driven	301 non-null	int64
5	Fuel_Type	301 non-null	object
6	Seller_Type	301 non-null	object
7	Transmission	301 non-null	object
8	Owner	301 non-null	int64

Output:

```
dtypes: float64(2), int64(3), object(4)
memory usage: 21.3+ KB
```

Check the number of missing values in each column

```
data.isnull().sum()
```

Output:

Column	0
Car_Name	0
Year	0
Selling_Price	0
Present_Price	0
Kms_Driven	0
Fuel_Type	0
Seller_Type	0
Transmission	0
Owner	0

Output:

```
dtype: int64
```

Show the number of rows and columns in the dataset

```
data.shape
```

Output:

```
(301, 9)
```

Count occurrences of each Fuel_Type in the dataset

```
data['Fuel_Type'].value_counts()
```

Output:

Fuel_Type	count
Petrol	239
Diesel	60
CNG	2

Output:
```
dtype: int64
```

Count occurrences of each Seller_Type in the dataset

```
data['Seller_Type'].value_counts()
```

Output:

Seller_Type	count
Dealer	195
Individual	106

Output:
```
dtype: int64
```

Count occurrences of each Transmission type in the dataset

```
data['Transmission'].value_counts()
```

Output:

Transmission	count
Manual	261
Automatic	40

Output:
```
dtype: int64
```

Step 4: Data Preprocessing

Encode Fuel_Type column by mapping categorical values to numerical (Petrol:0, Diesel:1, CNG:2)

```
data = data.replace({'Fuel_Type':{'Petrol':0, 'Diesel':1, 'CNG':2}})
data
```

Output:
```
Dataset after Fuel_Type Encoding:
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	0	Dealer	Manual	0
1	sx4	2013	4.75	9.54	43000	1	Dealer	Manual	0
2	ciaz	2017	7.25	9.85	6900	0	Dealer	Manual	0
3	wagon r	2011	2.85	4.15	5200	0	Dealer	Manual	0
4	swift	2014	4.60	6.87	42450	1	Dealer	Manual	0

...	...	...	...	...	...	...	...	...	...
296	city	2016	9.50	11.60	33988	1	Dealer	Manual	0
297	brio	2015	4.00	5.90	60000	0	Dealer	Manual	0
298	city	2009	3.35	11.00	87934	0	Dealer	Manual	0
299	city	2017	11.50	12.50	9000	1	Dealer	Manual	0
300	brio	2016	5.30	5.90	5464	0	Dealer	Manual	0

Output:

```
301 rows x 9 columns
```

Encode Seller_Type column by mapping categorical values to numerical (Dealer:0, Individual:1)

```
data = data.replace({'Seller_Type':{'Dealer':0, 'Individual':1}})
data
```

Output:

```
Dataset after Seller_Type Encoding:
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	0	0	Manual	0
1	sx4	2013	4.75	9.54	43000	1	0	Manual	0
2	ciaz	2017	7.25	9.85	6900	0	0	Manual	0
3	wagon r	2011	2.85	4.15	5200	0	0	Manual	0
4	swift	2014	4.60	6.87	42450	1	0	Manual	0
...	...	...	...	...	...	...	...	...	...
296	city	2016	9.50	11.60	33988	1	0	Manual	0
297	brio	2015	4.00	5.90	60000	0	0	Manual	0
298	city	2009	3.35	11.00	87934	0	0	Manual	0
299	city	2017	11.50	12.50	9000	1	0	Manual	0
300	brio	2016	5.30	5.90	5464	0	0	Manual	0

Output:

```
301 rows x 9 columns
```

Encode Transmission column by mapping categorical values to numerical (Manual:0, Automatic:1)

```
data = data.replace({'Transmission':{'Manual':0, 'Automatic':1}})
data
```

Output:

```
Dataset after Transmission Encoding:
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	ritz	2014	3.35	5.59	27000	0	0	0	0
1	sx4	2013	4.75	9.54	43000	1	0	0	0
2	ciaz	2017	7.25	9.85	6900	0	0	0	0
3	wagon r	2011	2.85	4.15	5200	0	0	0	0
4	swift	2014	4.60	6.87	42450	1	0	0	0
...	...	...	...	...	...	...	...	...	...
296	city	2016	9.50	11.60	33988	1	0	0	0

297	brio	2015	4.00	5.90	60000	0	0	0	0
298	city	2009	3.35	11.00	87934	0	0	0	0
299	city	2017	11.50	12.50	9000	1	0	0	0
300	brio	2016	5.30	5.90	5464	0	0	0	0

Output:

```
301 rows x 9 columns
```

Step 5: Model Training

Create feature matrix X by dropping Car_Name and Selling_Price columns

```
X = data.drop(['Car_Name','Selling_Price'], axis=1)
print("Feature matrix X created by dropping 'Car_Name' and 'Selling_Price' columns.")
```

Output:

```
Feature matrix X created by dropping 'Car_Name' and 'Selling_Price' columns.
```

Create target vector Y containing the Selling_Price column

```
Y = data['Selling_Price']
print("Target vector Y created from 'Selling_Price' column.")
```

Output:

```
Target vector Y created from 'Selling_Price' column.
```

Display the first 5 rows of the feature matrix X

```
X.head()
```

Output:

```
Feature Matrix (X) Preview:
```

Output:

	Year	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner
0	2014	5.59	27000	0	0	0	0
1	2013	9.54	43000	1	0	0	0
2	2017	9.85	6900	0	0	0	0
3	2011	4.15	5200	0	0	0	0
4	2014	6.87	42450	1	0	0	0

Display the first 5 values of the target vector Y

```
Y.head()
```

Output:

```
Target Vector (Y) Preview:
```

Output:

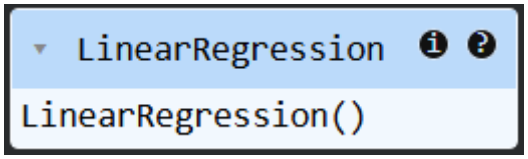
	Selling_Price
0	3.35
1	4.75
2	7.25
3	2.85
4	4.60

Output:
```
dtype: float64
```

 # Split the dataset into training and testing sets, then train a Linear Regression model

```
X_train, X_test, Y_train, Y_test = train_test_split(X, Y, test_size = 0.15, random_state =
    123)
model = LinearRegression()
model.fit(X_train, Y_train)
print("Model Trained Successfully")
```

Output:
```
Model Trained Successfully
```



Step 6: Model Evaluation

 # Generate predictions for training and testing datasets using the trained model

```
Y_train_pred = model.predict(X_train)
Y_test_pred = model.predict(X_test)
print("Predictions generated for both Train and Test sets.")
```

Output:
```
Predictions generated for both Train and Test sets.
```

 # Evaluate model performance on training data using MAE, MSE, RMSE, and R2 score

```
mae = mean_absolute_error(Y_train, Y_train_pred)
mse = mean_squared_error(Y_train, Y_train_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_train, Y_train_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root MSE (RMSE):", rmse)
print("R^2 Score:", r2)
```

Output:

```
Mean Absolute Error (MAE): 1.1605491562123338
Mean Squared Error (MSE): 3.015625977191586
Root MSE (RMSE): 1.7365557800403608
R^2 Score: 0.8869753499147778
```

Evaluate model performance on testing data using MAE, MSE, RMSE, and R2 score

```
mae = mean_absolute_error(Y_test, Y_test_pred)
mse = mean_squared_error(Y_test, Y_test_pred)
rmse = np.sqrt(mse)
r2 = r2_score(Y_test, Y_test_pred)

print("Mean Absolute Error (MAE):", mae)
print("Mean Squared Error (MSE):", mse)
print("Root MSE (RMSE):", rmse)
print("R^2 Score:", r2)
```

Output:

```
Mean Absolute Error (MAE): 1.3160017493663516
Mean Squared Error (MSE): 3.8164774430610064
Root MSE (RMSE): 1.953580672684903
R^2 Score: 0.8114116982277362
```

Retrieve the learned coefficients and intercept of the linear regression model

```
print("Coefficients:", model.coef_)
print("Intercept:", model.intercept_)
```

Output:

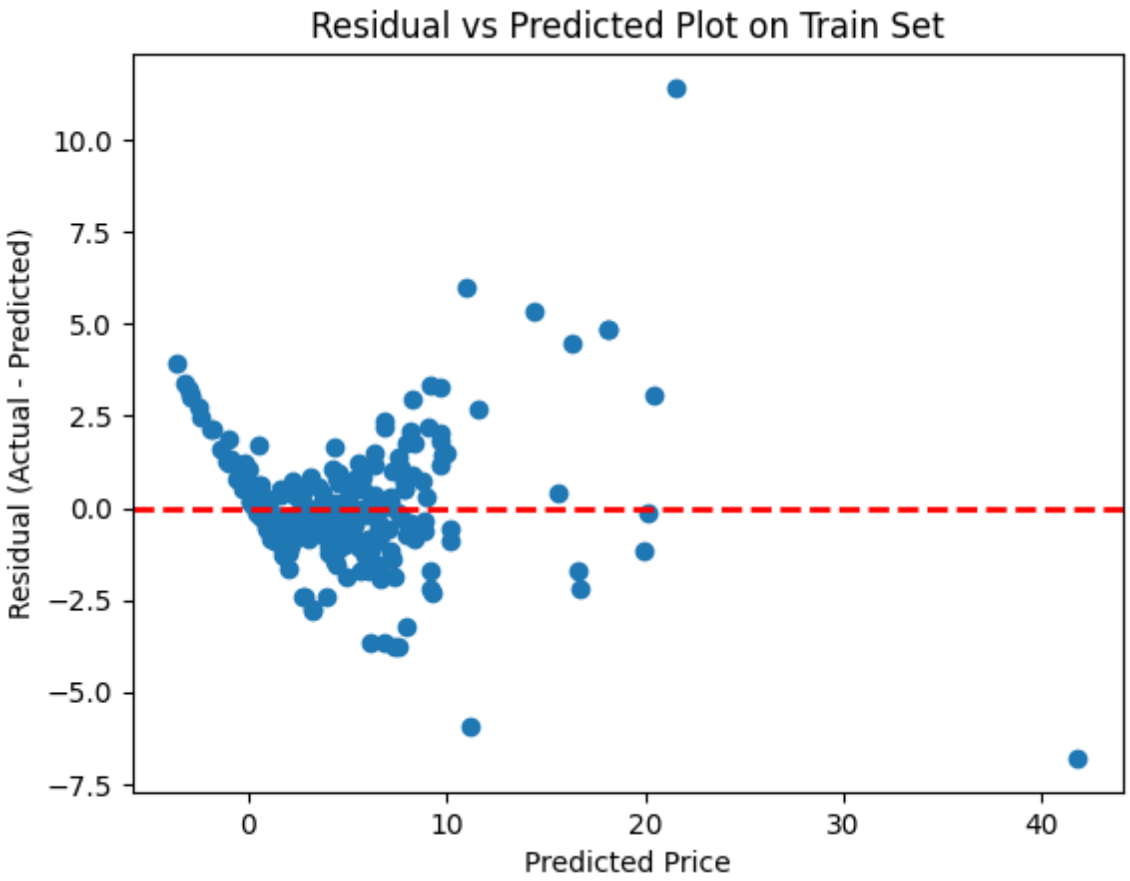
```
Coefficients: [ 3.92825483e-01  4.39836584e-01 -5.25747442e-06  1.38880051e+00 -1.25197058e+00
1.44799328e+00 -7.41998907e-01]
Intercept: -789.4879877881141
```

Plot residuals against predicted values on training data to analyze error patterns

```
residuals = Y_train - Y_train_pred
plt.scatter(Y_train_pred, residuals)
plt.axhline(0, color='red', linestyle='--', linewidth=2)
plt.xlabel("Predicted Price")
plt.ylabel("Residual (Actual - Predicted)")
plt.title("Residual vs Predicted Plot on Train Set")
plt.show()
```

Output:

```
Residual vs Predicted Plot on Train Set
```

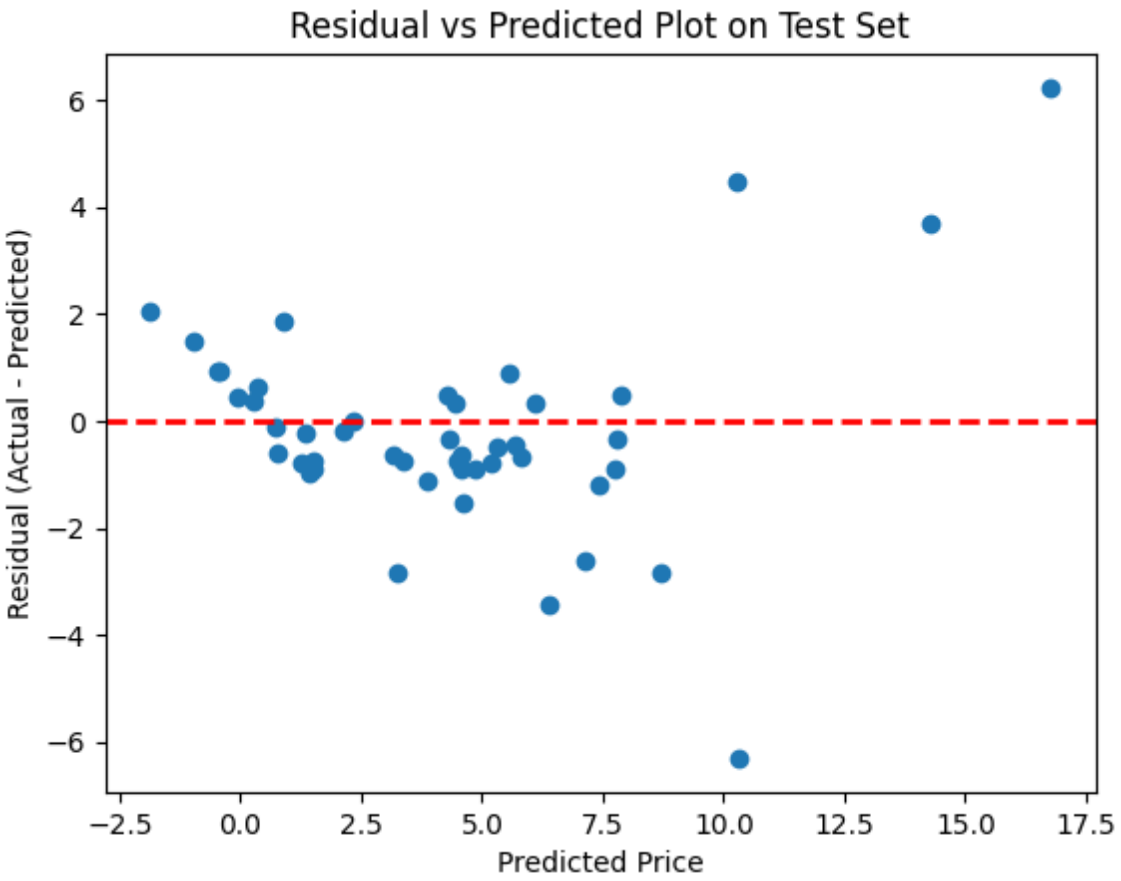


 # Plot residuals against predicted values on testing data to assess model generalization

```
residuals = Y_test - Y_test_pred
plt.scatter(Y_test_pred, residuals)
plt.axhline(0, color='red', linestyle='--', linewidth=2)
plt.xlabel("Predicted Price")
plt.ylabel("Residual (Actual - Predicted)")
plt.title("Residual vs Predicted Plot on Test Set")
plt.show()
```

Output:

```
Residual vs Predicted Plot on Test Set
```



 # Choose one random test sample from the dataset to compare the actual and predicted selling prices.

```
test_samples = data.sample(n=5, random_state=42)
display(test_samples)
print("Interactive Prediction Table Loaded")
```

Output:

```
Interactive Prediction Table Loaded
Choose one random test sample.
```

Output:

	Car_Name	Year	Selling_Price	Present_Price	Kms_Driven	Fuel_Type	Seller_Type	Transmission	Owner

---

Interactive Features (Static View)

Interactive Feature: Test Sample Predictions

Clicking a row in the test samples table shows the model prediction. Below are all 5 test samples with their prediction results:

Index	Car Name	Year	Actual	Present	Kms	Fuel	Seller	Trans	Owner	Predicted	Error
31	ritz	2011	2.35	4.89	54200	Petrol	Dealer	Manual	0	2.35	0.00
144	Bajaj Pulsar NS 200	2014	0.60	0.99	25000	Petrol	Individual	Manual	0	0.71	-0.11
42	sx4	2008	1.95	7.15	58000	Petrol	Dealer	Manual	0	2.15	-0.20
114	Royal Enfield 350	2015	1.15	1.47	17000	Petrol	Individual	Manual	0	1.36	-0.21
89	etios g	2014	4.75	6.76	40000	Petrol	Dealer	Manual	0	4.43	0.32